Ascertainment of non-coding genes as key molecular players in cervical squamous cell carcinoma through the systems biology approach

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Abstract

Background

Cervical cancer (CC) is the primary cause of mortality among women in developing countries. Preventing cervical cancer is partially possible by early vaccination against the human papillomavirus, the most common cause of the disease. Nevertheless, it is imperative to understand the genetics of the disease progression to develop new therapeutic strategies. With the advent of NGS technologies, a vast amount of biological data for cancers can be used to extract meaningful insights. The Cancer Genome Atlas project provides multi-omics profiling datasets for each type of cancer under the study. We performed integrative network analysis to explore competing endogenous RNAs (ceRNAs) as potential candidate drivers for cervical squamous cell carcinoma.

Objective

The present study aims to identify potential genes and pathways associated with cervical carcinoma progression through integrative networks and systems biology approach.

Methods

Differential gene expression patterns (i.e., protein-coding and non-coding genes) were filtered by differential expression analysis on transcriptomic profiling data of cervical cancer and normal cervix tissue. Dysregulated PPI network analysis on differentially expressed protein-coding genes results in key hub genes regulating the network. The key hub genes were validated by the SVM model. However, non-coding genes play a significant role in regulating protein-coding genes resulting in the progression of the disease and hence the interactions between the key genes and their target miRNA and lncRNA were studied with the help of integrative network analysis. Further, functional enrichment and survival analysis were performed to understand their role in disease progression.

Results

In this study, we analyzed cancer-specific expression networks, i.e., differentially expressed lncRNA and their target microRNA network & differentially expressed microRNA and their target hub mRNA networks which were identified via differential expression and gene co-expression network analysis on transcriptomic data of cervical cancer. Subsequent regulatory network analysis on each network separately identifies key genes in the network that have a crucial role in regulating the network, i.e., regulation of disease progression. It is found that microRNA "hsa-miR-184" is present in both target networks and involved in several types of cancer by playing a role in regulating gene expression. Network analysis of DElncRNA – target DEmiRNA network shows that KCNQ1OT1, as a hub long non-coding RNA plays an important regulatory role in transcription of target genes via epigenetic modifications. Also, DEmiRNA – hub DEmRNA regulatory network analysis results in hsa-miR-637 as hub micro-RNA has a role in posttranscriptional regulation of target gene expression. Further, the competing endogenous RNAs (ceRNAs) network clearly shows the relationship between lncRNAs and miRNAs. These genes are mostly enriched in biological processes of cell division, mitotic nuclear division, cell cycle checkpoint and cell proliferation in gene ontology analysis. The KEGG pathway enrichment analysis of the proteins lists them as mainly associated with the cell cycle. We also performed a survival analysis to check the potential candidate's prognosis for their overall survival in cervical cancer patients.

Conclusions

Our findings provide key insights into the regulatory role of lncRNA and miRNA via cancer-specific expression networks in the pathogenesis of cervical cancer. Further, they can be validated for the development of diagnostic markers and therapeutic studies.

Keywords: Cervical cancer, Non-coding genes, ceRNA, Candidate drivers, Regulatory networks and Support vector machine learning algorithm